MSG No. 99.170 P_I4mm [Type IV, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4mm`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `2mm.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,4,5,6]`
2	$\left[\frac{1}{2},\ 0,\ z\right]$	`[2,3,7,8]`
3	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[9,12,13,14]`
4	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[10,11,15,16]`

Table 3: Wyckoff site: `8c`, site symmetry: `..m`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,8]`
2	$[-x,\ x,\ z]$	`[2,5]`
3	$[x,\ -x,\ z]$	`[3,6]`
4	$[-x,\ -x,\ z]$	`[4,7]`
5	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,16]`
6	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[10,13]`
7	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[11,14]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[12,15]`

Table 4: Wyckoff site: `8d`, site symmetry: `.m.`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,6]`
2	$[0,\ x,\ z]$	`[2,8]`
3	$[0,\ -x,\ z]$	`[3,7]`
4	$[-x,\ 0,\ z]$	`[4,5]`
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,14]`
6	$\left[\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[10,16]`
7	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[11,15]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[12,13]`

Table 5: Wyckoff site: `16e`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-y,\ x,\ z]$	`[2]`

continued ...

Table 5

No.	position	mapping
3	$[y, -x, z]$	[3]
4	$[-x, -y, z]$	[4]
5	$[-x, y, z]$	[5]
6	$[x, -y, z]$	[6]
7	$[-y, -x, z]$	[7]
8	$[y, x, z]$	[8]
9	$\left[x+\frac{1}{2}, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[9]
10	$\left[\frac{1}{2}-y, x+\frac{1}{2}, z+\frac{1}{2}\right]$	[10]
11	$\left[y+\frac{1}{2}, \frac{1}{2}-x, z+\frac{1}{2}\right]$	[11]
12	$\left[\frac{1}{2}-x, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[12]
13	$\left[\frac{1}{2}-x, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[13]
14	$\left[x+\frac{1}{2}, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[14]
15	$\left[\frac{1}{2}-y, \frac{1}{2}-x, z+\frac{1}{2}\right]$	[15]
16	$\left[y+\frac{1}{2}, x+\frac{1}{2}, z+\frac{1}{2}\right]$	[16]